

Research paper

Topographic fragmentation shapes maternal lineage divergence and plastome variation in the seed dispersal limited alpine herb *Rhodiola bupleuroides*

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ABSTRACT

The fragmented landscape of the Qinghai-Xizang Plateau (QXP) is widely recognized as a driver of speciation. However, the relative contributions of physical topographic barriers versus environmental heterogeneity in shaping the fine-scale genetic structure of alpine plants remain debated. Particularly for species with limited seed dispersal, population plastomic evidence remains limited regarding the extent to which topographic fragmentation restricts seed-mediated gene flow and contributes to maternal lineage differentiation. Here, we employed *Rhodiola bupleuroides* as a model system, sequencing whole-chloroplast genomes of 468 individuals from 19 populations across its known range to investigate the relative roles of geographic isolation and environmental heterogeneity in shaping plastome genetic structure. Phylogenetic and molecular clock analyses recovered four maternal lineages, indicating that intraspecific divergence initiated at 3.88 Ma [2.83–5.31 Ma], with subsequent lineage splits extending into the Early Quaternary. Spatial genetic analyses showed a stronger association of plastome genetic structure with geographic isolation than with environmental heterogeneity (IBD > IBE; geographic distance: Mantel $r = 0.578$, $P = 0.0001$; environmental distance: Mantel $r = 0.275$, $P = 0.0001$). The major genetic discontinuities were spatially aligned with the mountain-canyon-river configuration of the QXP and adjacent mountains, suggesting that topographic barriers provide a primary spatial template for the long-term restriction of seed-mediated maternal gene flow. Beyond spatial isolation, lineage-specific plastome variation was also evident: the high-diversity Hengduan Mountain lineage was characterized by elevated chloroplast DNA diversity and enriched complex SSR motifs, whereas the southwestern plateau lineage combined stronger codon usage bias with lower nucleotide diversity and reduced retention of complex SSR motifs. These contrasts indicate that geographically isolated lineages have embarked on distinct plastome evolutionary trajectories associated with their specific topographic and habitat contexts. Together, these results suggest that while topographic fragmentation and restricted dispersal primarily shape maternal lineage divergence, environmental heterogeneity and lineage-specific demographic history further shape fine-scale plastome variation. This

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study provides high-resolution plastome-based evidence for geography-associated maternal lineage divergence in a seed-limited alpine herb on the QXP.

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1. Introduction

The Qinghai–Xizang Plateau (QXP) and its southeastern fringe, the Hengduan Mountains, are major centers of alpine plant diversity shaped by complex topography, climatic gradients, and late Cenozoic landscape reorganization (Nagy and Grabherr, 2009; Spicer et al., 2020; Liu et al., 2021; Yu et al., 2024). While the region's extraordinary species richness is closely associated with topographic and climatic evolution, we have yet to fully grasp how specific landscape-scale processes shape fine-scale genetic patterns (Badgley et al., 2017). For lineages occupying high-alpine niches, rugged topography creates physical barriers and heterogeneous microenvironments along steep elevational gradients (Cao et al., 2025a). This makes the QXP an appropriate system for evaluating the relative roles of geographic isolation and environmental heterogeneity in lineage differentiation (Yu et al., 2019; Li et al., 2025b). Against this backdrop, although wind-pollinated trees in this region have been well studied, we still lack empirical insights into how the maternal lineages of short-range alpine herbs persist across deep valleys and high ridges—especially through the integrative lens of high-resolution maternal markers across a range-wide scale (Tang et al., 2024, 2025). In insect-pollinated alpine herbs, gene flow is expected to be asymmetric: pollen-mediated dispersal may be less restricted across ridges, whereas seed-mediated dispersal is more directly constrained by deep valleys and dissected terrain (Cruzan and Hendrickson, 2020; Lewis et al., 2023). Such pollen–seed asymmetry predicts that biparentally inherited nuclear markers may be partially homogenized by long-distance pollen flow, while maternally inherited plastid genomes retain sharper geographic discontinuities that track historical colonization routes and topographic resistance (Xu et al., 2021; Fichant et al., 2023). Therefore, plastome-based phylogeography provides a hypothesis-driven approach for evaluating seed-mediated maternal lineage structure in complex mountain landscapes (Cao et al., 2025b; Liu et al., 2025b). Notably, much of the QXP's canyon–ridge architecture and monsoon reorganization intensified since the late Cenozoic, providing a plausible long-term topographic and climatic background on which later Quaternary dynamics may have been superimposed (Ding et al., 2022; Cai et al., 2023; He et al., 2024; Ahmad et al., 2025). This leaves us with an open question: is the genetic differentiation we observe primarily associated with spatial isolation and physical barriers, or is it additionally shaped by environmental heterogeneity after accounting for spatial structure?

At the heart of deciphering speciation within complex terrains lies the strategic choice of genetic markers capable of untangling historical dispersal pathways (de Brito, 2025). The plastid genome (plastome) stands out here, defined by its uniparental inheritance, lack of recombination, and relatively conserved evolutionary rate (Camus et al., 2022; Cao et al., 2025b; Liu et al., 2025a; Yadav et al., 2025). While the nuclear genome integrates signals from both pollen and seed flow, the plastome—typically maternally inherited, largely tracking seed-mediated dispersal—offers a useful marker system for reconstructing maternal lineage history (Scott, 2024; Chung, 2025). Importantly, this framework motivates a falsifiable prediction for future nuclear data: range-wide nuclear

SNPs (biparental markers) may show weaker topographic breaks (or more admixture) than plastomes because pollen-mediated gene flow can homogenize biparental genomes across barriers (Hu and Ennos, 1999; Xu et al., 2021). Such cytonuclear discordance—arising from pollen–seed asymmetry, incomplete lineage sorting, or organellar capture—has become increasingly recognized in phylogeographic and phylogenomic studies (Folk et al., 2017; Lee-Yaw et al., 2019; Larson et al., 2026). Framing plastome analyses in this context allows organellar variation to be interpreted as the maternal component of landscape genetic structure (Doyle, 2022; Li et al., 2025a). Beyond reconstructing dispersal history, plastomes can also provide information on lineage-specific coding and microstructural variation, although such signals require cautious interpretation in the absence of population-level nuclear genomic data (Greiner et al., 2011; Jarvis and López-Juez, 2013; Sierra et al., 2023). If long-term isolation reduces effective population size, plastomes may accumulate lineage-specific mutational signatures (e.g., SSR motifs) via drift, while localized environmental conditions may be associated with codon usage variation (Chu and Wei, 2021; Müller et al., 2022; Yang et al., 2023b). Thus, the plastome functions as a marker system that records maternal demographic history and lineage-specific plastome variation in heterogeneous mountain habitats (Kurata et al., 2024). Recent advances in sequencing have enabled population plastomics, allowing researchers to resolve subtle genetic variation that traditional markers typically overlook (Dodsworth, 2015; Twyford and Ness, 2017; De Coster et al., 2021). Topographically complex and environmentally buffered habitats can promote long-term lineage persistence, population subdivision, and the retention of lineage-specific plastome variation (Finocchiaro et al., 2023; Keppel et al., 2024). Accordingly, our intraspecific analyses focus on seed-mediated maternal lineage history captured by plastomes, while broader nuclear context is explored using complementary genus-level analyses (Song et al., 2025).

Testing these hypotheses hinges on identifying a model system that combines specific life-history traits with a strategic distribution pattern. An appropriate system should occupy high-elevation habitats, span major topographic units of the QXP, and possess life-history traits that make seed-mediated dispersal sensitive to terrain fragmentation (Manel et al., 2003; Holderegger and Wagner, 2008; Wang, 2013; Gueuning et al., 2017). *Rhodiola bupleuroides* (Wall. ex Hook. f. & Thomson) S.H. Fu (Crassulaceae) matches these requirements. Biologically, *R. bupleuroides* is expected to experience a “double constraint” on dispersal: insect pollination and a lack of obvious specialized seed-dispersal structures (e.g., pappus-like appendages), making seed movement largely gravity-biased and locally retained (Zhu and Lou, 2010; Cheplick, 2022; Tuthill et al., 2023). This life-history profile may enhance topography-driven restriction of seed flow and render maternal lineages particularly sensitive to canyon–ridge fragmentation, potentially providing finer phylogeographic resolution than wind-pollinated conifers, where long-distance pollen flow can dilute spatial structure (Chybicki and Oleksa, 2018; Kling and Ackerly, 2021; Xu et al., 2021). The species occurs in alpine shrublands and rocky crevices, and its distribution extends from

the eastern margins to the plateau interior, including the deeply fragmented Hengduan Mountains (Zhang et al., 2015; You et al., 2018, 2022; Spicer et al., 2020; Yang et al., 2023a; Li et al., 2026). Compared with heavily harvested congeners such as *R. crenulata* (Hook. f. & Thoms.) H. Ohba, *R. bupleuroides* appears less affected by direct human exploitation (Xin et al., 2015; Fu et al., 2017; Tao et al., 2019; Zhuang et al., 2019). This less confounded evolutionary background makes it suitable for examining how habitat fragmentation and environmental heterogeneity have jointly shaped maternal lineage divergence and plastome variation across the Pliocene to Early Quaternary interval (Epps and Keyghobadi, 2015; Favre et al., 2015; Xing and Ree, 2017; Wan et al., 2018).

Despite the recognized importance of the QXP as a biodiversity hotspot, a critical question remains regarding the mechanistic role of its rugged terrain: Does complex topography act merely as a physical barrier that impedes movement, or does it actively contribute to lineage-specific plastome variation by isolating lineages across contrasting habitats? This distinction is particularly pertinent for alpine herbs with limited seed dispersal, which are theoretically most susceptible to landscape fragmentation. Using *Rhodiola bupleuroides* as a representative model, we evaluated whether topographic fragmentation acts not only as a geographic barrier, but also as a spatial template within which restricted seed-mediated dispersal, environmental heterogeneity, and demographic history are associated with maternal lineage structure and lineage-specific plastome variation. In this study, we assembled the whole-chloroplast genomes of 468 individuals from 19 populations, encompassing the known range of *R. bupleuroides*. We combined this population-scale plastome dataset with a complementary genus-level Angiosperms353 analysis across representative *Rhodiola* species. The genus-level analysis provided a broader nuclear phylogenetic and environmental context by assessing phylogenetic signals of environmental variables and comparing candidate environment-related gene trees with the nuclear species tree using RF distances. Against this broader context, the intraspecific plastome dataset was used to address three main questions: (1) When did the major maternal lineages diverge, and how does their timing fit within the late Cenozoic landscape history of the QXP and adjacent mountains? (2) Is plastome genetic structure more strongly associated with geographic isolation than with environmental heterogeneity after accounting for spatial structure? (3) Do maternal lineages occupying contrasting Hengduan and plateau habitats differ in plastome diversity, SSR profiles, and codon usage patterns?

2. Materials and methods

2.1. Material collection and sequencing

A total of 468 individuals from 19 natural populations were collected across the known distribution range of *Rhodiola bupleuroides* (Table S1). The number of individuals sampled per population ranged from 15 to 30. The sampling localities spanned 27.0013–31.8919°N and 86.0372–101.0383°E, covering the main geographic regions occupied by this species, including the Hengduan Mountains, the eastern margin of the QXP, the plateau interior, and the southwestern plateau margin. The sampled populations covered an east–west geographic span of approximately 1469.56 km and a north–south span of approximately 823.16 km, with pairwise geographic distances among populations ranging from approximately 15.93–1469.56 km. They also represented a broad elevational gradient from 3200 to 5200 m a.s.l., covering the main topographic and environmental conditions occupied by this species. Total genomic DNA was extracted from silica-gel dried leaves using the CTAB method (Li et al., 2013). For

all 468 individuals, paired-end libraries with a fragment size of approximately 300–450 bp and an insert size of approximately 350–400 bp were constructed and sequenced as 150 bp paired-end reads using the BGI-DIPSEQ platform (Huang et al., 2017). In addition, Nanopore long-read sequencing was performed for one representative individual using a GridION X5 sequencer and the SQK-LSK108 library preparation kit (Cherf et al., 2012). This individual was selected based on high molecular-weight DNA quality, sufficient DNA quantity, and its suitability for long-read validation of plastome structural continuity, circularity, repeat-associated regions, and IR/SC junction boundaries.

2.2. Chloroplast genome assembly and annotation

Raw data underwent quality control using Trimmomatic v.0.36 (Bolger et al., 2014) to generate clean reads, employing the following parameters: ILLUMINACLIP:TruSeq3-PE.fa:2:35:4:12:true, LEADING:3, TRAILING:3, SLIDINGWINDOW:5:15, and MINLEN:50. The BGI-DIPSEQ short-read clean reads from each individual were used as the primary input for chloroplast genome assembly using GetOrganelle v.1.7 (Jin et al., 2020), with multiple k-mer values of 45, 65, 85, 95 and 115. The same multi-k-mer setting was applied to all samples during the GetOrganelle assembly process. Assembly and annotation metrics for all 468 plastomes, including total plastome length, LSC, SSC, and IR lengths, regional GC contents, and gene counts, are provided in Table S12. Samples with uncertain circularization or ambiguous IR/SC junctions and repeat regions were further checked or assembled following the four-step procedure of Dong et al. (2022), which included chloroplast-related contig identification, contig extension and connection, validation of IR/SC junctions, and confirmation of circular plastome structure. Nanopore long reads from the representative individual were used as auxiliary evidence to validate plastome structural continuity, circularity, repeat-associated regions, and IR/SC junction boundaries. The lengths of different plastid regions and their GC contents were calculated using custom Python scripts. Complete plastid sequences were annotated using the PlastidHub-PGA v.2.0 (Qu et al., 2019; Zhang et al., 2025b), and all annotations were subsequently manually curated in Geneious Prime 2025.2 (<https://www.geneious.com>). Finally, the circular plastid map of *R. bupleuroides* was visualized using TBtools-II v.2.450 (Chen et al., 2023).

2.3. Chloroplast genome variation analysis

All 468 chloroplast genomes were initially aligned using MAFFT v.7.526 (Katoh and Standley, 2013), followed by manual inspection and adjustment in MEGA X (Kumar et al., 2018) to correct misaligned regions. Genomic variations within the *Rhodiola bupleuroides* plastomes, specifically SNPs and InDels, were identified using snp-sites v.2.5.1 (Page et al., 2016) and DnaSP 6 (Rozas et al., 2017). To characterize genomic microstructure, Simple Sequence Repeats (SSRs) were identified using CPStools (Huang et al., 2024) with the following parameters: mononucleotides ≥ 10 , dinucleotides ≥ 5 , trinucleotides ≥ 4 , and tetra- to hexanucleotides ≥ 3 . The abundance and distribution of these variations (SNPs, InDels, and SSRs) across genomic regions were analyzed and visualized using custom Python scripts and TBtools-II v.2.450 (Chen et al., 2023). Furthermore, polymorphism information content (PIC) values were calculated to assess sequence divergence hotspots.

2.4. Phylogenetic inference and population structure analyses

Phylogenetic relationships were inferred from the complete plastome alignment using the Maximum Likelihood (ML) method implemented in IQ-TREE 2 (Minh et al., 2020), with *Phedimus*

aizoon (L.) 't Hart selected as the outgroup because *Phedimus* is closely related to, but outside, *Rhodiola* within Crassulaceae and has been used as an outgroup in previous *Rhodiola* phylogenetic studies. The resulting phylogenetic trees were visualized and refined using FigTree v.1.4.2 (<http://beast.bio.ed.ac.uk/FigTree>) and iTOL v.7 (Letunic and Bork, 2024). Population genetic structure was analyzed based on an intraspecific chloroplast SNP matrix using ADMIXTURE v.1.3.0 (Alexander and Lange, 2011). SNPs extracted from the complete plastome alignment were filtered using PLINK v.1.9 (Purcell et al., 2007) with thresholds of minor allele frequency (MAF) ≥ 0.01 , variant missing rate ≤ 0.10 , and individual missing rate ≤ 0.10 . Of the 5053 raw variant sites, 2311 high-quality SNPs and all 468 individuals passed filtering and were retained for ADMIXTURE and PCA analyses. We performed analyses for a range of K values from 2 to 10. The optimal number of clusters (K) was determined by identifying the lowest cross-validation (CV) error or the distinct inflection point of the error curve. The population structure corresponding to the optimal K was visualized using the R package "ggplot2" (Wickham, 2011). Additionally, Principal Component Analysis (PCA) was conducted using PLINK v.1.9 (Purcell et al., 2007) based on the same 2311-SNP matrix, and the results were visualized using custom Python scripts. To assess the robustness of phylogenetic inference, a supplementary SNP-based ML tree was reconstructed using 2268 SNPs from the outgroup-included dataset under the same filtering criteria.

2.5. Haplotype network and genetic differentiation analyses

Haplotypes were identified and their frequencies within populations were determined using NetST (<https://github.com/sculab/NetST>). A haplotype network was subsequently constructed using the minimum spanning network (MSN) method implemented in NetST. Pairwise fixation index (F_{ST}) values between different sub-lineages were estimated using VCFtools v.0.1.17 (Danecek et al., 2011), while nucleotide diversity (π) and haplotype diversity (Hd) were calculated using DnaSP 6 (Rozas et al., 2017).

2.6. Divergence time estimation and demographic history inference

Divergence times were estimated using MCMCTree (Yang and Rannala, 2006; Rannala and Yang, 2007) as implemented in GeneMiner2 (<https://github.com/sculab/GeneMiner2>) (Zhang et al., 2022; Xie et al., 2024; Yu et al., 2026), under an independent-rates model. For molecular dating, we used an expanded plastome dataset including the 468 *R. bupleuroides* individuals and eight external reference taxa: *Rhodiola alternata*, *R. chrysanthemifolia*, *R. gelida*, *R. himalensis*, *R. purpureoviridis*, *R. sinuata*, *Phedimus aizoon*, and *P. kamtschaticus*. Among these reference taxa, plastome sequences of *P. aizoon*, *R. chrysanthemifolia*, *R. himalensis*, and *R. purpureoviridis* were obtained from our sequencing data (He et al., 2026), whereas those of *R. alternata*, *R. gelida*, *R. sinuata*, and *P. kamtschaticus* were retrieved from publicly available NCBI plastome sequences. Detailed source information and accession numbers are provided in Table S2. Three secondary calibration constraints from Ren et al. (2023) were applied: (1) the stem age of *Rhodiola* (11.45–26.12 million years ago; Ma), rooted using *P. aizoon* and *P. kamtschaticus* (Crassulaceae) outside the focal genus; (2) the divergence between the clade containing *R. alternata*, *R. chrysanthemifolia*, and *R. sinuata* and the remaining *Rhodiola* lineages included in this analysis (4.09–9.61 Ma); and (3) the divergence between the clade containing *R. gelida*, *R. himalensis*, and *R. purpureoviridis* and the *R. bupleuroides* lineage group (3.60–8.30 Ma). Demographic history was reconstructed using Bayesian Skyline Plots (BSP) implemented in BEAST v.2.7.8 (Bouckaert et al., 2014) under the HKY + Γ substitution model with four gamma categories and an uncorrelated lognormal relaxed

molecular clock. The MCMC chain was run for 20 million generations, with parameters and trees sampled every 1000 generations. The first 10% of posterior samples were discarded as burn-in. Convergence was assessed in Tracer, and effective sample size (ESS) values > 200 were considered acceptable. The relaxed-clock BSP results were used for the final demographic reconstruction of each cluster. Additionally, neutrality tests (Tajima's D and Fu's F_s) (Tajima, 1989; Fu, 1997) were performed in DnaSP 6 (Rozas et al., 2017) using the complete plastome alignment for each cluster to detect signals of population expansion or contraction.

2.7. Landscape genetics and spatial barrier detection

To visually identify geographic barriers to maternal gene flow, we applied Monmonier's algorithm in Barrier v.2.2 (Manni et al., 2004). A 19×19 population genetic distance matrix was calculated as the mean pairwise TN93 distance among populations from the complete plastome alignment using the *dist.dna* function in the R package *ape*, and then mapped onto a Delaunay triangulation constructed from sampling coordinates. This population-level sequence divergence matrix was used to visualize major spatial genetic discontinuities in Barrier. We identified the first four barriers ($K = 4$) to visualize major genetic discontinuities superimposed on the topography.

To assess IBD and IBE, we integrated climate, soil, elevation, and human footprint data (Table S3). Climate data and elevation-related layers were obtained from the WorldClim database (<https://www.worldclim.org/>), the ENVIREM environmental raster database (<http://envirem.github.io/>), and CGIAR-CSI (<https://cgiarcsi.community>). Soil variables were obtained from the Harmonized World Soil Database v.2.0 through the FAO soil portal (<https://www.fao.org/>), with a spatial resolution of 1 km. Human impact variables, including the human influence index and human footprint data, were obtained from the Center for International Earth Science Information Network (CIESIN), Columbia University (<http://www.ciesin.columbia.edu>). Variables were harmonized for resolution and projection, excluding those with constant values. To minimize multicollinearity, environmental variables were screened using the Variance Inflation Factor (VIF), and variables with VIF values below 5.0 were retained for downstream analyses. This filtering retained 12 environmental variables for construction of the environmental distance matrix, and the retained variables and their VIF values are listed in Table S4. IBD and IBE were assessed using Mantel and partial Mantel tests (999 permutations). Genetic distances for the IBD and IBE analyses were derived from pairwise IBS values calculated from the filtered SNP dataset using PLINK (Purcell et al., 2007), while geographic and environmental distances were calculated using great-circle and Euclidean metrics, respectively.

Environmental associations were assessed in Python using the IBS-based genetic distance matrix and environmental distance matrices described above. Partial Mantel tests (controlling for geographic distance) were performed for each of the 12 VIF-retained environmental variables (Pearson's r ; 999 permutations; Benjamini–Hochberg correction, $q < 0.05$). Multivariate effects were quantified using distance-based Redundancy Analysis (dbRDA) (Legendre and Anderson, 1999). The first 20 PCoA axes of the IBS-based genetic distance matrix, which jointly explained 99.34% of the IBS-based genetic distance variation (Table S11D), were used as the response matrix, with standardized environmental variables as predictors (999 permutations).

2.8. Codon usage and coding-sequence divergence analyses

To evaluate lineage-level variation in plastome coding sequences, Codon Usage Bias (CUB) was analyzed using CPStools

(Huang et al., 2024). We calculated the Relative Synonymous Codon Usage (RSCU) and the Effective Number of Codons (ENC). ENC values, ranging from 20 (extreme bias) to 61 (no bias), were used to quantify the strength of codon usage bias across different lineages. Pairwise Ka/Ks ratios were also estimated from concatenated chloroplast CDS alignments using the model-averaging method in KaKs_Calculator 3.0 (Zhang, 2022) following codon-based alignment. Low-quality alignments and invalid pairs ($K_a = 0$, $K_s = 0$, or extreme K_s) were excluded. Because gene-level Ka/Ks estimates were generally uninformative for many plastid genes due to the limited number of synonymous and non-synonymous substitutions, the concatenated-CDS Ka/Ks results were used only as an exploratory overview of broad coding-sequence divergence, rather than as a formal gene-level test of positive selection. Statistical analyses and visualization of CUB and coding-sequence divergence results were performed using custom Python scripts.

2.9. Genus-level nuclear phylogenetic signal and candidate environment-related gene analyses

To provide a genus-level nuclear context for the environmental patterns observed in *Rhodiola bupleuroides*, we conducted complementary analyses across representative *Rhodiola* species. Angiosperms353 genes were recovered from genome skimming data of 19 representative species (He et al., 2026), including 18 *Rhodiola* species and *Phedimus aizoon* as the outgroup, using GeneMiner2 (Zhang et al., 2022; Xie et al., 2024; Yu et al., 2026), with the Angiosperms353 reference set as queries. The reference set was obtained from the *Rhodiola rosea* dataset ERR7621166 and contained 326 Angiosperms353-derived target genes. After GeneMiner2 target-gene quality evaluation, 293 high-quality genes were selected for recovery. No additional fixed identity or target-coverage thresholds were applied after GeneMiner2 recovery; recovered sequences were compared with the reference sequences to summarize the observed pairwise identity and reference coverage (Table S19). Sampling information and data sources for these species are provided in Table S5. The recovered sequences were aligned with MAFFT v.7.526 (Katoh and Standley, 2013), trimmed to remove poorly aligned regions, and used to infer single-gene trees. In total, 292 single-gene trees were successfully inferred and used to reconstruct the Angiosperms353-derived nuclear species tree, with *P. aizoon* as the outgroup.

Species distribution records for *Rhodiola* were obtained from GBIF and supplemented with available literature records and field collection information (Table S6). Records with uncertain coordinates, duplicated localities, or obvious geographic errors were removed. Environmental variables were extracted for the filtered records and summarized at the species level using the mean, standard deviation, 5th percentile, and 95th percentile (Table S7). These summaries, together with the 353-gene nuclear species tree, were imported into RASP 4.1 (Yu et al., 2015, 2020) for phylogenetic signal analysis using Pagel's λ and Blomberg's K . Variables with $P < 0.05$ were considered significant, and variables with $0.05 \leq P < 0.10$ were treated as marginally significant (Table S8).

Candidate environment-related genes were selected from the annotated gene set of the published *Rhodiola yunnanensis* genome (Wang et al., 2026). Based on functional annotations, 53 candidate genes related to temperature response, drought or water-stress response, light or UV response, oxidative stress, and phenylpropanoid-related pathways were retained. These genes were used as queries in GeneMiner2 to recover homologous sequences from the same genome skimming data. For each recovered gene, sequences were aligned with MAFFT v.7.526 (Katoh and Standley, 2013), trimmed, and used to infer a single-gene tree.

Normalized Robinson–Foulds (RF) distances (Robinson and Foulds, 1981) between each single-gene tree and the 353-gene nuclear species tree were calculated using custom Python scripts, and genes with normalized RF distance ≤ 0.6 were retained as candidate genes with relatively high topological congruence to the nuclear species tree. The threshold was used to screen gene trees with relatively low topological distance from the species tree for subsequent exploratory comparison. Candidate gene information and RF distances are listed in Table S9.

3. Results

3.1. Identification of population structure and phylogenetic framework

Drawing on a high-fidelity SNP dataset from whole-plastome alignments of 468 individuals, we performed a systematic characterization of the population genetic structure in *Rhodiola bupleuroides*. The ADMIXTURE and PCA analyses were based on the same intraspecific SNP matrix containing 2311 high-quality chloroplast SNPs. The main ML phylogeny was inferred from the complete plastome alignment, whereas a supplementary SNP-based ML tree was reconstructed using 2268 SNPs obtained from the outgroup-included dataset under the same filtering criteria. Our ADMIXTURE-based inference, guided by cross-validation (CV) error analysis, showed that while error values generally trended downward, the rate of decline leveled off notably after $K = 4$ (Fig. S1). Benchmarking this against the clustering patterns at $K = 3$ and $K = 5$ (Fig. S2) suggested that $K = 4$ captures the most biologically and geographically coherent resolution, establishing it as our optimal model. Under this four-cluster framework, the populations sorted into distinct, well-supported geographic units (Fig. 1A and B): Cluster A occupies the transition between the eastern plateau margin and the Hengduan Mountains; Cluster B spreads across the Hengduan core; Cluster C anchors the region extending from the central plateau interior to the eastern margin; and Cluster D remains concentrated along the southwestern fringe of the QXP (Fig. 1A).

To unravel the evolutionary trajectory and phylogenetic relationships among the four identified clusters, we reconstructed a Maximum Likelihood (ML) tree using whole-plastome sequences. A supplementary SNP-based ML tree reconstructed from 2268 SNPs obtained from the outgroup-included dataset under the same filtering criteria showed a topology broadly congruent with the whole plastome phylogeny (Fig. S3). The resulting topology (Fig. 1C) supported that *Rhodiola bupleuroides* has split into four well-supported monophyletic lineages, showing a pattern consistent with our ADMIXTURE clustering. Within this phylogenetic scaffold, Cluster A was the first to diverge, positioning itself as sister to the rest of the ingroup. This initial split was followed by the emergence of the Cluster B lineage. Finally, Cluster C and Cluster D were resolved as sister groups; as the most recently diverged terminal lineages (Fig. 1C), they exhibit the highest degree of genetic affinity within the species.

Principal Component Analysis (PCA) bolstered the reliability of our population groupings, projecting 468 individuals into a multi-dimensional genetic space and unearthing the fine-scale nuances between sister groups C and D. Overall, the first two axes captured a substantial portion of the variation, with PC1 and PC2 accounting for 23.24% and 18.93% of the total genetic variance, respectively. Cluster A segregated sharply along the negative pole of PC1, underscoring its pronounced genetic distinctiveness, while Cluster B occupied an independent space along the negative axis of PC2 (Fig. 1D). Notably, Clusters C and D exhibited partial convergence within the PC1–PC2 plane, which is consistent with their recent

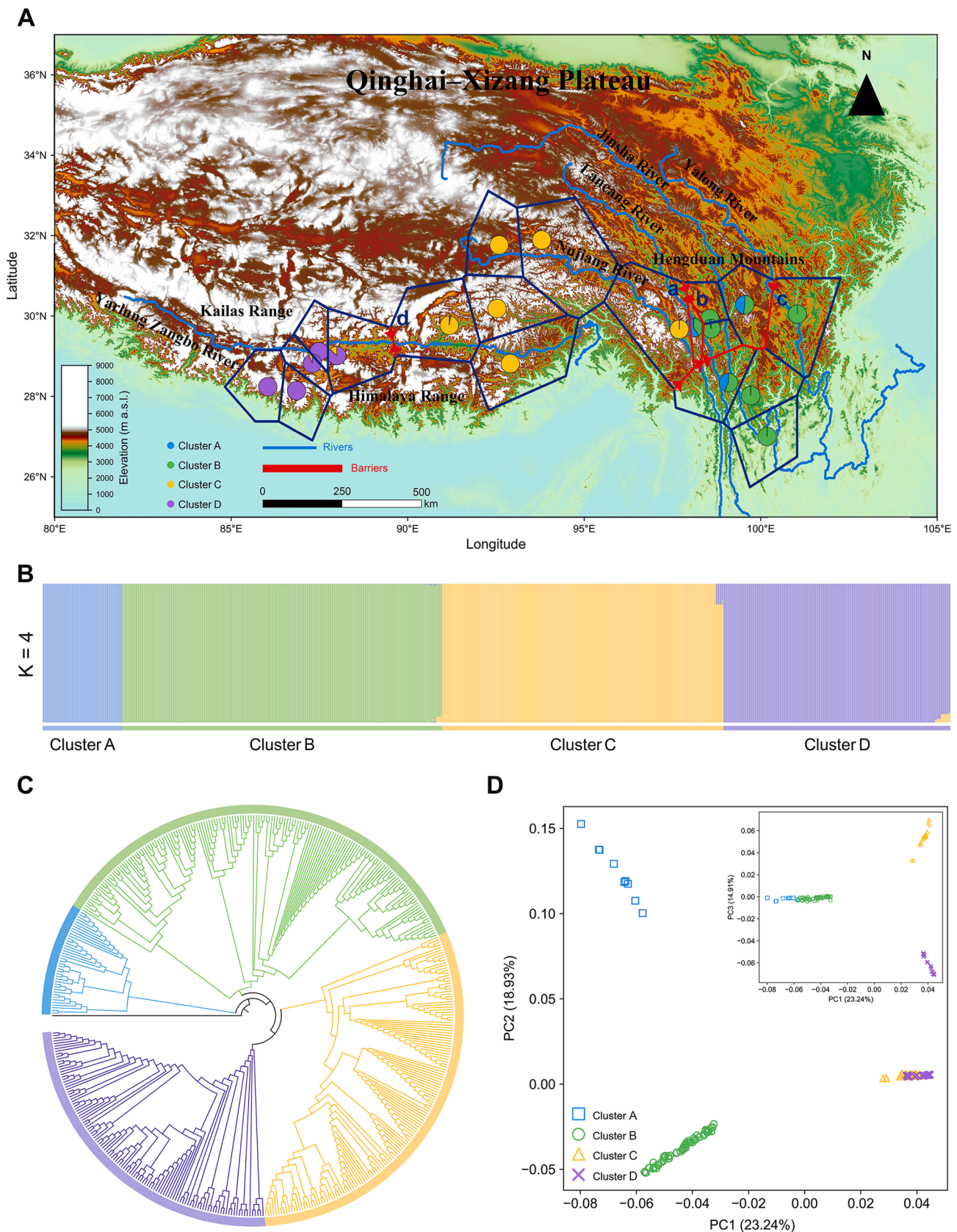


Fig. 1. Maternal genetic structure and geographic barriers of *Rhodiola bupleuroides* populations. (A) Geographic distribution of 19 natural populations with admixture proportions based on ADMIXTURE analysis. The background map indicates elevation gradients in meters above sea level (m a.s.l.). Pie charts represent the genetic composition of each population, colored according to the four identified genetic clusters (Clusters A–D; blue, Cluster A; green, Cluster B; yellow, Cluster C; purple, Cluster D). Blue polygons represent the Voronoi tessellation, and thick red lines (labeled a–d) indicate significant genetic barriers inferred by Monmonier's algorithm. Blue lines indicate major rivers. Major

divergence and sister-group relationship in the ML phylogeny. To further distinguish these closely related lineages, we incorporated the third principal component (PC3, 14.91% variance explained), which provided additional lineage-discriminating information not fully captured by the first two axes (Fig. 1D, top-right inset). The PC1–PC3 projection showed a clear separation, with Cluster C and Cluster D shifting toward the positive and negative ends of PC3, respectively. Further explorations across PC2–PC3 and PC3–PC4 (Fig. S4) supported the separation of these two closely related lineages and the overall independence of the four clusters in multidimensional genetic space.

3.2. Haplotype diversity and patterns of genetic differentiation

To refine our understanding of the internal diversity and maternal connections within the four clusters, we scaled our analysis to the haplotype level. From the complete chloroplast genome alignments of 468 individuals, we recovered 177 distinct haplotypes and visualized their relationships through a minimum spanning network (Fig. 2A). The spatial and lineage-specific partitioning of these haplotypes was uneven among lineages: Cluster B had the highest haplotype richness, harboring 71 haplotypes, followed by Cluster D (56) and Cluster C (38). In contrast, Cluster A was represented by 12 haplotypes, pointing toward a much more constricted genetic reservoir in this lineage.

The haplotype network (Fig. 2A) clearly delineated four groupings in high alignment with our ADMIXTURE and PCA results, signaling a robust pattern of genetic differentiation. While inter-cluster connections persist, each group maintains a dense internal architecture consistent with lineage-specific haplotype structure. Notably, network topologies differed among lineages. Cluster D displayed a relatively star-like pattern, whereas Cluster B was characterized by a complex, knot-like reticulate network, with more reticulate internal connections. In contrast, the sparser frameworks of Clusters A and C showed comparatively simpler haplotype configurations. To provide descriptive environmental context for these maternal lineages, we projected the mean elevation of sampling sites onto the node interiors using a grayscale gradient—shading from white (relatively lower elevations within the sampled range) to black (relatively higher elevations within the sampled range). The grayscale shading in Fig. 2A and the elevation summary in Fig. 2B show how sampling-site elevations are distributed across the four clusters and provide background information for the subsequent environmental association analyses. Furthermore, the ML tree derived from haplotype sequences (Fig. 2D) displayed high topological congruence with the whole-plastome phylogeny, further supporting the monophyly and divergence sequence of these maternal lineages—a hierarchy that closely mirrors the broad geographic gradient from the eastern margin through the Hengduan Mountains to the plateau interior.

Across the four lineages (Fig. 2C), nucleotide diversity (π) and haplotype diversity (H_d) showed contrasting patterns. Cluster B had the highest nucleotide diversity ($\pi = 0.00173$) and high haplotype diversity ($H_d = 0.969$), whereas Cluster D showed the lowest nucleotide diversity ($\pi = 0.00028$) but retained high haplotype diversity ($H_d = 0.973$) and 56 haplotypes. This indicates that Cluster D contains many distinct but closely related haplotypes, with limited average sequence divergence among them. Pairwise F_{st} values—spanning an interval from 0.64795 to

0.84507—indicated substantial plastome differentiation among clusters. Specifically, the highest differentiation was observed between Clusters A and D ($F_{st} = 0.84507$), whereas even the lowest-differentiated pair, Clusters A and B, still maintained substantial differentiation at 0.64795. Clusters C and D were resolved as sister lineages in the plastome-based ML tree, but they also showed high pairwise F_{st} , indicating that their plastome variant or haplotype frequencies have diverged markedly after lineage separation.

3.3. Lineage divergence history and population dynamics

Molecular dating via the MCMCTree chronogram (Fig. 2D) dated the split between the *Rhodiola bupleuroides* lineage group and its closest external reference taxa to 4.45 Ma (95% HPD: 3.60–5.94 Ma). Within *R. bupleuroides*, the onset of intraspecific maternal lineage divergence was estimated at 3.88 Ma (95% HPD: 2.83–5.31 Ma), corresponding to the separation of Cluster A from the remaining lineages. Cluster B subsequently diverged from the common ancestor of Clusters C and D at 3.43 Ma (95% HPD: 2.39–4.70 Ma), followed by the split between the sister lineages Cluster C and Cluster D at 2.49 Ma (95% HPD: 1.50–3.60 Ma). The within-cluster crown divergences were estimated at 1.85 Ma (Cluster A; 95% HPD: 0.81–2.99 Ma), 2.50 Ma (Cluster B; 95% HPD: 1.52–3.62 Ma), 1.45 Ma (Cluster C; 95% HPD: 0.73–2.34 Ma), and 1.11 Ma (Cluster D; 95% HPD: 0.48–1.88 Ma). These results indicate a Pliocene to Early Quaternary framework for the major maternal lineage divergences in *R. bupleuroides*.

The demographic histories of the four genetic clusters across the QXP reveal lineage-specific trajectories. By integrating Tajima's D , Fu's F_s , and Bayesian Skyline Plots (BSP), we found that the effective population size (N_e) of Cluster A (Fig. 2H) exhibited long-term stasis for nearly 2 Ma, showing no evidence of major expansions or contractions. This demographic stability is mirrored by its non-significant Tajima's D (−0.24133) and Fu's F_s (49.172) (Table 1). In contrast, Cluster B (Fig. 2G) showed an increase after approximately 2 Ma and has since sustained a relatively large population size, a pattern consistent with its high haplotype richness with 71 haplotypes recovered. For Cluster C, the N_e remained largely stable overall, with only a slight increase within the last ~0.5 Ma, which aligns with its marginally significant Tajima's D (−1.61746). Cluster D showed a relative signal of demographic increase, with the main rise in N_e occurring around ~1 Ma (Fig. 2E). Its Tajima's D was the most negative among the four clusters (−2.08417, $P < 0.05$), although Fu's F_s did not indicate a correspondingly strong expansion signal. Together, these results suggest a moderate and lineage-specific demographic increase in Cluster D.

3.4. Spatial barriers and stronger IBD than IBE in plastome genetic structure

To evaluate the relative associations of geographic isolation and environmental heterogeneity with plastome genetic structure, we first examined the spatial correspondence between maternal lineages and major topographic features. The spatial footprint of *Rhodiola bupleuroides* showed clear geographic correspondence with major topographic features of the QXP and adjacent mountains (Fig. 1A). Leveraging Monmonier's algorithm via Barrier v.2.2—integrating Delaunay triangulation with a TN93 distance

mountain ranges and river systems are labeled for geographic reference. (B) Bar plot of population genetic structure estimated by ADMIXTURE for $K = 4$. Each vertical bar represents an individual, and colors represent the proportion of inferred ancestral lineages. (C) Maximum Likelihood (ML) phylogenetic tree based on complete plastome sequences. Branch colors correspond to the genetic clusters defined in panel B. (D) Principal Component Analysis (PCA) plot of the 468 individuals. The main panel shows PC1 versus PC2, and the inset shows PC1 versus PC3.

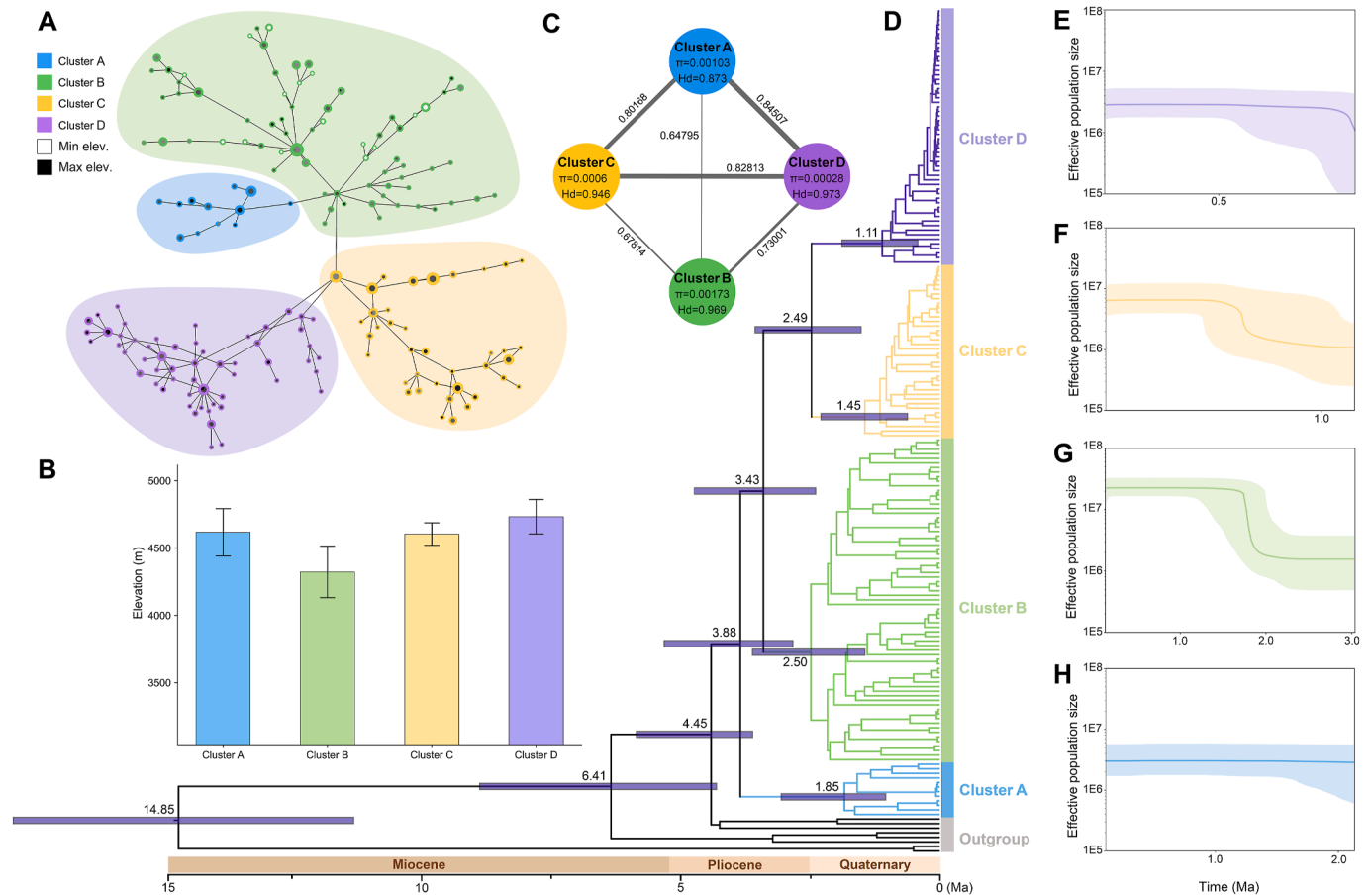


Fig. 2. Phylogeographic structure and evolutionary history of *Rhodiola bupleuroides*. (A) Haplotype network showing genealogical relationships, with inner shading representing relative elevation within the sampled range from minimum elevation (white) to maximum elevation (black). Circle size is proportional to haplotype frequency, and colors indicate the four genetic clusters: blue, Cluster A; green, Cluster B; yellow, Cluster C; purple, Cluster D. (B) Bar plot summarizing the elevation distribution of sampling sites across the four identified clusters. Bars indicate mean elevation, and error bars indicate variation among sampling sites within each cluster. (C) Schematic of genetic differentiation (F_{st}) and diversity indices (π , H_d) among the four identified clusters. Numbers along connecting lines indicate pairwise F_{st} values; π and H_d represent nucleotide diversity and haplotype diversity, respectively. (D) Time-calibrated phylogenetic tree estimated using MCMCTree, with 95% HPD intervals shown as purple bars. Node labels indicate divergence times in Ma. (E–H) Demographic history reconstruction using Bayesian Skyline Plots (BSP) for Cluster D, Cluster C, Cluster B, and Cluster A, respectively. The plots trace the median effective population size (N_e) and 95% HPD intervals (shaded) over time.

Table 1

Genetic diversity and neutrality tests based on chloroplast genome sequences for the four clusters dataset of *Rhodiola bupleuroides*.

Cluster	N	S	H	Tajima's <i>D</i>	Fu's <i>F_s</i>
A	41	686	12	-0.24133	49.172
B	165	2853	71	-1.67317	53.442
C	145	945	38	-1.61746	41.858
D	117	584	56	-2.08417	0.176

matrix—we identified four major genetic discontinuities that were spatially associated with mountain ranges, deep valleys, and major river systems (Fig. 1A). Cluster A occurred mainly in the transition zone between the eastern QXP margin and the Hengduan Mountains, a terrain characterized by steep elevational gradients and prominent topographic scarps. This physical isolation corresponded to the distinct geographic boundary of this early-diverging lineage. Cluster B was distributed mainly in the Hengduan Mountains, where the landscape is intricately carved by deep-seated river systems, notably the Jinsha, Lancang, and Nu. These deep canyon systems coincided with the highest haplotype richness and with multiple internal genetic discontinuities within this high-diversity lineage. The emergence of multiple sub-

clusters here (Barriers b and c in Fig. 1A) suggests that high-relief terrain has contributed to restricted maternal connectivity within the Hengduan region. Cluster C occupies the threshold between the central plateau platforms and the southeastern margin. Here, sharp elevational shifts and deep-dry valleys form a major topographic transition zone (Barrier a in Fig. 1A) that is spatially associated with the genetic discontinuity separating Cluster C from adjacent lineages. Finally, Cluster D is mainly distributed within the Himalaya–Gangdise ranges along the plateau's southwestern rim. It is partitioned from Cluster C by the upper Yarlung Tsangpo valley and towering northern ranges (Barrier d in Fig. 1A). Despite their relative spatial proximity, these topographic features coincide with marked plastome differentiation between the sister lineages C and D.

To gauge the restrictive influence of these geographic barriers on gene flow, we performed a series of Mantel tests. The environmental distance matrix was constructed using the 12 variables retained after VIF filtering (Table S3). A significant positive correlation emerged between genetic and geographic distances (Mantel $r = 0.578$, $P = 0.0001$; Fig. 3B–D), indicating a strong broad-scale association between geographic isolation and plastome genetic differentiation in *Rhodiola bupleuroides*. While environmental distance also correlated with genetic divergence, the association

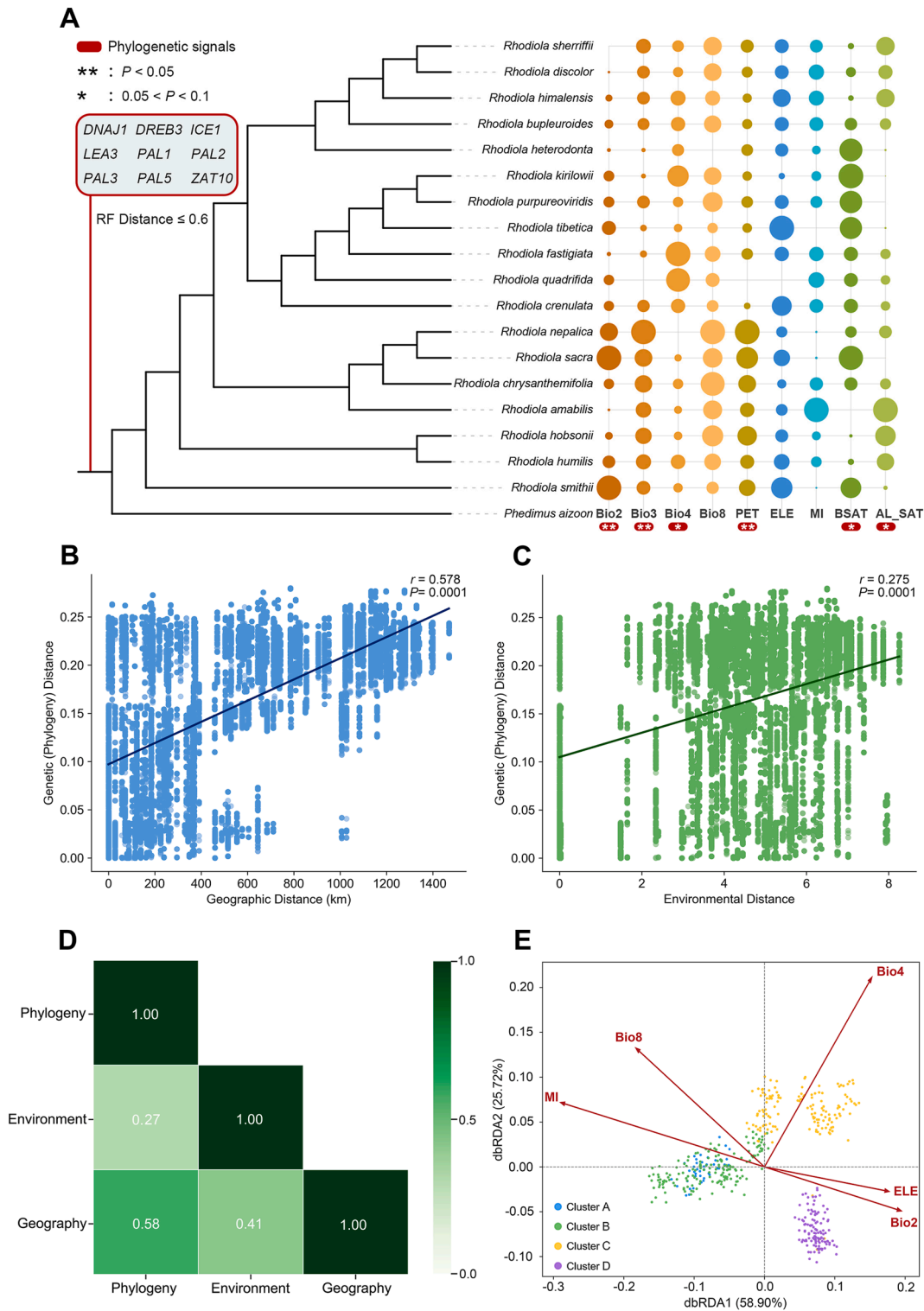


Fig. 3. Relative effects of geographic isolation and environmental heterogeneity on plastome genetic structure in *Rhodiola bupleuroides*. (A) Genus-level nuclear phylogenetic signals and candidate environment-related genes in *Rhodiola*. The tree is based on Angiosperms353-derived nuclear genes. Bubble size represents species-level environmental values. Red marks indicate environmental variables with phylogenetic signals; double asterisks indicate $P < 0.05$, and single asterisks indicate $0.05 < P < 0.1$. The red boxed genes are candidate environment-related genes with normalized RF distance ≤ 0.6 relative to the nuclear species tree. RF, Robinson–Foulds distance; ELE, elevation; MI, moisture index; PET, potential evapotranspiration; BSAT, base saturation; AL_SAT, aluminium saturation. (B) Isolation by distance (IBD): Scatter plot showing the significant positive correlation between phylogenetic (genetic) distance and geographic distance ($r = 0.578$, $P = 0.0001$). The fitted line indicates the Mantel correlation based on 999 permutations. (C) Isolation by environment (IBE): Scatter plot showing the correlation between phylogenetic distance and environmental distance ($r = 0.275$, $P = 0.0001$). The fitted line indicates the Mantel correlation based on 999 permutations. (D) Heatmap of Pearson correlation coefficients among phylogenetic, environmental, and geographic distances. Darker green indicates stronger correlations. (E) Distance-based Redundancy Analysis (dbRDA) ordination plot. Colored dots represent individuals from different genetic clusters (A–D; blue, Cluster A; green, Cluster B; yellow, Cluster C; purple, Cluster D). Red arrows indicate the direction and magnitude of the most explanatory VIF-retained environmental variables (e.g., MI, Bio8, Bio4, ELE, and Bio2). MI, moisture index; Bio8, mean temperature of the wettest quarter; Bio4, temperature seasonality (standard deviation $\times 100$); ELE, elevation; Bio2, mean diurnal range. Percentages on dbRDA axes indicate the proportion of constrained variation explained by each axis.

was noticeably less pronounced (Mantel $r = 0.275$, $P = 0.0001$), indicating that environmental heterogeneity also contributed to genetic structure, but with a smaller effect size than geographic distance. Partial Mantel tests further supported this pattern: the link between genetic and geographic distances remained robust even after partitioning out environmental effects ($r = 0.531$, $P = 0.0001$). Conversely, the residual correlation between genetics and environment—though statistically significant—yielded a much smaller effect size once spatial factors were accounted for ($r = 0.051$, $P = 0.0057$; Fig. S5A and B). Collectively, these results indicate that spatial genetic structure is primarily associated with geographic isolation, while environmental heterogeneity contributes a weaker but detectable independent signal.

3.5. Environmental correlates of plastome genetic differentiation and genus-level nuclear context

Having established a stronger broad-scale association between geographic isolation and plastome genetic structure, we next examined whether environmental heterogeneity retained detectable signals within this geographically structured background. To place these environmental dimensions in a broader nuclear phylogenetic context in *Rhodiola*, we summarized genus-level phylogenetic signals for selected environmental variables across representative *Rhodiola* species. We mapped selected environmental variables onto the 353-gene nuclear species tree (Fig. 3A and Table S7). Gene recovery was high across the ingroup *Rhodiola* species, with an average of 291.39 ± 0.92 recovered genes per species, corresponding to $99.45\% \pm 0.31\%$ of the 293 GeneMiner2-selected high-quality target genes (Table S19). Phylogenetic signals were detected for several variables, mainly related to temperature variability, potential evapotranspiration, and soil properties. Among the variables shown in Fig. 3A, Bio2, Bio3, and PET showed significant K-based phylogenetic signals. Additional marginal signals were detected for Bio4, BSAT, and ALUM_SAT. Elevation, moisture index (MI), and Bio8 were also shown to facilitate comparison with the population-level landscape genetic results, although they did not show strong K-based phylogenetic signals in this analysis. The complete Pagel's λ and Blomberg's K results are provided in Table S8.

To identify environmental variables associated with genetic differentiation while minimizing the confounding influence of geographic distance, we performed single-variable partial Mantel tests between genetic distance and each of the 12 VIF-retained environmental variables, using geographic distance as the control matrix. Results summarized in Table S10 showed that several variables retained significant but generally weak independent associations with genetic distance. The strongest positive associations were detected for MI ($r = 0.181$, $P = 0.001$), Bio2 ($r = 0.157$, $P = 0.001$), ELE ($r = 0.090$, $P = 0.001$), and Bio4 ($r = 0.046$, $P = 0.001$). In contrast, Bio8 and GYPSUM showed the strongest negative associations ($r = -0.074$ and -0.078 , respectively; $P = 0.001$). These results indicate that environmental effects remain detectable when geographic distance was included as the control matrix, although their effect sizes were relatively limited.

We further used dbRDA to quantify the multivariate contribution of the VIF-retained environmental variables to genetic variation (Table S11). The overall dbRDA model was significant ($F = 0.6468$, $P = 0.001$). The first two constrained axes explained 58.90% and 25.72% of the constrained variation, respectively (Figs. 3E and S6). These percentages refer to the constrained variation within the environmental model, rather than the proportion of total plastome genetic variation explained by environmental variables. Among the retained variables, MI, Bio8, Bio4, Bio2, and ELE showed the highest explanatory power, explaining

25.20%, 13.81%, 12.69%, 11.84%, and 11.75% of the variation in the variable-wise dbRDA tests, respectively ($P = 0.001$ for all). Although the ranking of variables differed between single-variable partial Mantel tests and dbRDA, both analyses highlighted moisture-, temperature-, and elevation-related gradients as important environmental correlates of plastome genetic structure. Spatial projections of representative environmental gradients further illustrated the geographic heterogeneity of MI, Bio8, and Bio2 across the distribution range of *Rhodiola bupleuroides* (Figs. S7–S9). Overall, these results support a secondary but detectable role of environmental heterogeneity, superimposed on the stronger effect of geographic isolation.

To further complement this genus-level nuclear context, the RF distance calculation showed that nine candidate environment-related gene trees had normalized RF distance ≤ 0.6 relative to the 353-gene nuclear species tree, indicating relatively similar topologies between these gene trees and the species tree (Fig. 3A and Table S9). These genes included *DNAJ1*, *DREB3*, *ICE1*, *LEA3*, *PAL1*, *PAL2*, *PAL3*, *PAL5*, and *ZAT10*. Among them, *PAL5* showed the lowest RF distance, followed by *DNAJ1*. These genes were therefore retained as candidate environment-related genes whose single-gene trees showed relatively high topological congruence with the nuclear species tree.

After characterizing the spatial and environmental correlates of plastome genetic structure and their broader genus-level nuclear context, we examined whether the identified maternal lineages differed in plastome structural features, microstructural variation, and coding-sequence characteristics.

3.6. Plastome structural conservation and microstructural variation among lineages

Our assembly and annotation of 468 *Rhodiola bupleuroides* chloroplast genomes revealed a highly conserved architecture, characterized by the classic quadripartite structure common to angiosperm plastids (Fig. 4A). Each genome comprises a large single-copy (LSC) region and a small single-copy (SSC) region separated by a pair of inverted repeats (IRa/IRb), with total lengths ranging narrowly between 150,866 and 152,026 bp (averaging 151,577 bp). This structural stasis is further reflected in the GC content, which remained steady at approximately 37.70% across all individuals (Table S12 and Fig. S10). We identified 113 unique genes (131 in total), with gene order and intron architecture showing no deviation across the dataset. The absence of any significant rearrangements provided a consistent plastome framework for comparing sequence variation and microstructural features among lineages.

Our comprehensive alignment yielded 2311 high-quality single nucleotide polymorphisms (SNPs) alongside 48 insertion/deletion (InDel) events (Tables S13 and S14). Upon closer inspection of the filtered SNPs, we found a transition/transversion (Ti/Tv) ratio of 1.18, with transitions, most notably C $\rightarrow$ T and A $\rightarrow$ G, slightly outnumbering transversions (Fig. S11A). These InDel events spanned a range of 1–13 bp, though single-base shifts were by far the most prevalent (Fig. S11C). The genomic landscape of these variations is uneven across plastome regions (Fig. S11B). The Large Single-Copy (LSC) region contained the largest proportion of variants, hosting 69.7% of the SNPs and roughly 68% of the InDels. In stark contrast, the Inverted Repeat (IR) regions remained more conserved, contributing a mere 8% to the total variation. Functionally, the vast majority of these variants (over 86%) are clustered within exons and intergenic spacers (IGS), while introns remained relatively stable islands in the genome (Fig. S11D).

To further characterize the fine-scale distribution of genetic variation across the plastome, and to identify hypervariable loci

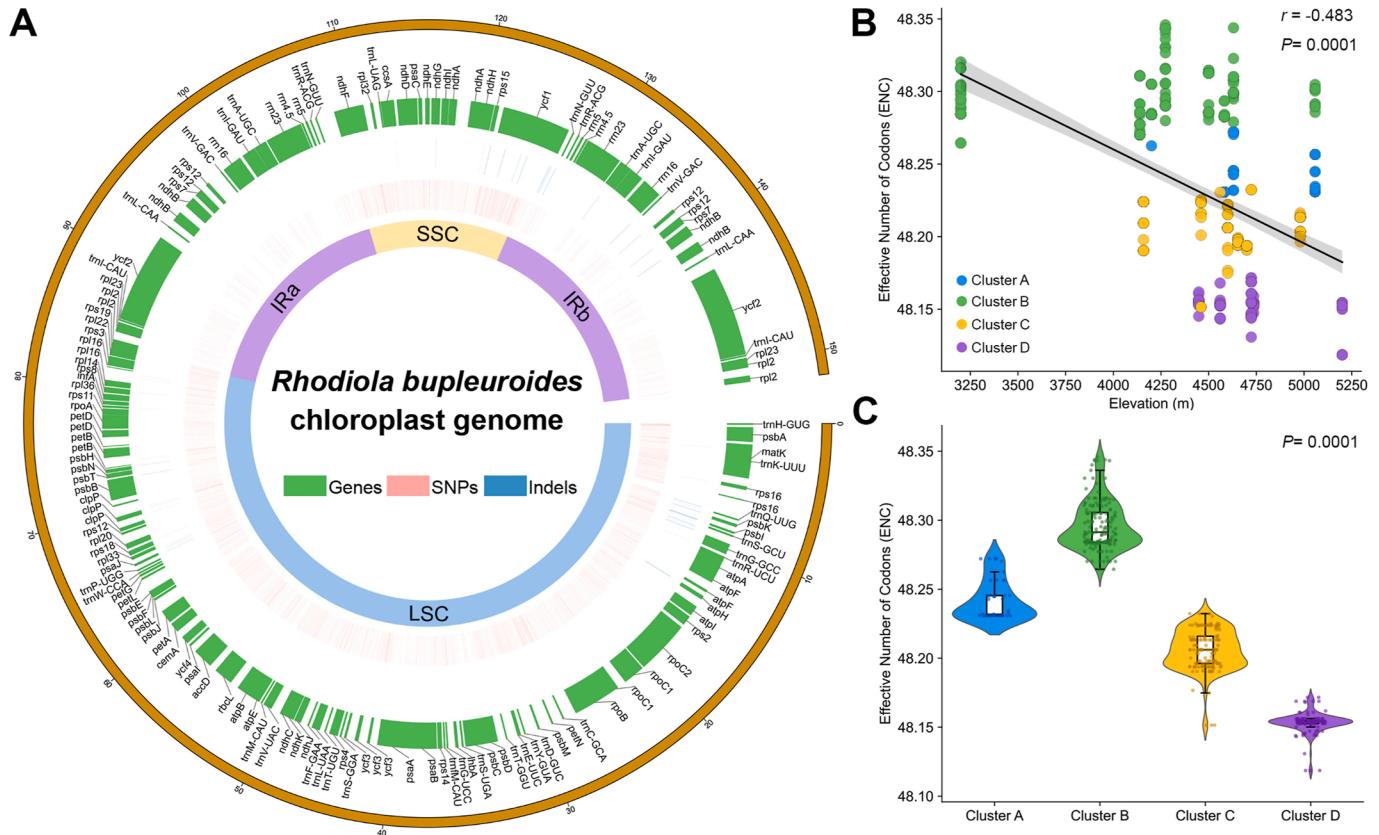


Fig. 4. Plastome variation and codon usage patterns in *Rhodiola bupleuroides*. (A) Circular map visualizing genome structure, gene annotation, and the density of SNPs and Indels across LSC, SSC, and IR regions. Green blocks indicate annotated genes, pink marks indicate SNPs, and blue marks indicate Indels. LSC, large single-copy region; SSC, small single-copy region; IR, inverted repeat region. (B) Scatter plot showing the negative correlation between codon usage bias (ENC) and elevation, colored by genetic cluster. The black line indicates the fitted regression, and the grey band indicates the confidence interval; $r = -0.483$ and $P = 0.0001$. Lower ENC values indicate stronger codon usage bias. (C) Violin plot showing the distribution of ENC values among the four genetic clusters. Inner boxplots and points show within-cluster variation; $P = 0.0001$ indicates significant differences among clusters.

within the LSC-dominated high-variation regions described above, we calculated the polymorphism information content (PIC) for each locus and pinpointed several hypervariable hotspots (Table S15). Among the 215 regions assessed, a subset of coding sequences (including *ycf1*, *rpoC2*, *ndhF*, *ycf2*, and *matK*) and intergenic spacers (e.g., *trnS-trnG*, *rps16-trnQ*, *rpoB-trnC*, *psbK-psbI*, and *ndhF-rpl32*) exhibited exceptionally high nucleotide polymorphism. Beyond highlighting rapidly evolving sectors of the plastome, these hypervariable sites provide a valuable resource for developing high-resolution molecular markers for future population genetic studies.

The pervasive A/T bias within these plastomes provides a background for examining simple sequence repeat (SSR) variation as a component of plastome microstructure. Our analysis of whole-genome SSR patterns (Table S16) showed that Cluster B had the highest SSR richness (53.88 ± 3.00), with relatively more tri- and penta-nucleotide complex motifs—features that are nearly absent in other lineages. These complex SSR motifs were mainly located within introns (mean: 11.82). By comparison, Cluster D showed a different SSR distribution pattern: although its total SSR count was intermediate, SSRs were mainly located in non-coding intergenic spacers (IGS; mean: 29.64), with few SSRs in coding regions or complex motifs. This contrast indicates lineage-associated differences in plastome microstructural variation between Clusters B and D, with Cluster B showing greater enrichment of complex SSR motifs and Cluster D showing a simpler SSR profile dominated by intergenic spacers.

3.7. Codon usage variation and exploratory coding-sequence divergence

To evaluate lineage-level variation in codon usage patterns, we began by quantifying codon usage bias (CUB) across the entire genome. Initial RSCU analysis (Table S17) showed that all lineages adhere to a classic angiosperm preference pattern, with optimal codons predominantly ending in A or T. Yet, set against this shared background, ENC quantification uncovered detectable differences between lineages (Fig. 4B and C; Table S18). Cluster D, mainly distributed in high-elevation plateau habitats, showed the lowest mean ENC values (48.15 ± 0.008), indicating relatively stronger codon usage bias. In contrast, Cluster B exhibited the highest ENC values (48.29 ± 0.017), indicating weaker codon usage bias among the four clusters. To further characterize these differences, we compared specific codon usage profiles (Fig. S12A), which showed that Cluster D was associated with a particular suite of preferred codons (GCT, GGA, GAT, GTA, TTA, and TCA). ENC values were negatively correlated with elevation (Fig. 4B), indicating that codon usage bias varied along the elevational gradient within the sampled range.

We also summarized broad coding-sequence divergence using concatenated chloroplast CDS Ka/Ks estimates, which are provided in Fig. S12B. This exploratory overview showed that elevated Ka/Ks comparisons occurred mainly within Cluster B, providing additional context for lineage-level coding-sequence divergence.

4. Discussion

4.1. High resolution of whole plastomes in tracing seed-mediated gene flow

Our results support a plastome-based view in which topographic fragmentation provides the spatial context for seed-mediated maternal lineage differentiation in *Rhodiola bupleuroides*. Angiosperm plastomes are often described as structurally conserved in gene order and overall architecture; nevertheless, our population-level data from *R. bupleuroides* point to a structure–sequence decoupling, where architectural conservation co-occurs with substantial nucleotide- and microstructural variation (Howe et al., 2003; Wicke et al., 2011; Daniell et al., 2016). Despite a rigid gene order and canonical quadripartite architecture, we identified 2311 high-quality SNPs, together with detectable InDels and SSR polymorphisms across populations. Variants were mainly located in the LSC region and concentrated in loci widely recognized as plastome hypervariable, such as *ycf1* and *ndhF* (Dong et al., 2015; Zhu et al., 2016; Amar, 2020). This pattern helps explain why commonly used short plastid markers may provide limited resolution for shallow intraspecific divergence, whereas complete plastome datasets can capture a broader spectrum of maternal lineage variation (Dong et al., 2015; Ruhssam et al., 2015). In the apparent absence of large-scale rearrangements, nucleotide substitutions and microstructural shifts (including SSR length variation) likely constitute the dominant sources of plastome-scale microevolutionary signal in this species.

Because plastomes are predominantly maternally inherited in angiosperms, they provide a direct window into seed-mediated dispersal that is less affected by pollen-mediated gene flow in the nuclear genome (Xu et al., 2021; Sakamoto and Takami, 2024; Chung, 2025). In the topographically dissected QXP, life-history traits that limit long-distance seed dispersal can amplify seed-mediated structure, while pollen-mediated gene flow may still reduce nuclear differentiation—creating the conditions for cytonuclear contrasts (Tackenberg, 2003; Zhou et al., 2024). A useful contrast comes from QXP conifers, where marker systems with different inheritance modes reveal strong asymmetry: in *Picea crassifolia* Kom., seed-mediated structure is pronounced while pollen-mediated gene flow is comparatively efficient, yielding much weaker geographic differentiation for the paternally inherited plastid marker (Meng et al., 2007). Thus, relative to nuclear markers that integrate both pollen and seed movement, plastome variation can more directly reflect seed-mediated colonization histories in dissected landscapes. Conceptually, this pattern is consistent with landscape-genetic expectations that rugged terrain increases effective resistance to gene flow, thereby “filtering” realized dispersal and generating strong spatial genetic structure (McRae, 2006; Manel and Holderegger, 2013).

The high inter-population divergence ($F_{st} = 0.65\text{--}0.85$) observed here is consistent with expectations under strong seed-dispersal limitation in dissected terrain, a scenario that should elevate differentiation in uniparentally inherited markers relative to biparental nuclear loci (Ennos, 1994). This mechanism may also explain why Clusters C and D, although sister lineages in the plastome-based ML tree, nevertheless show high pairwise F_{st} : after their separation at approximately 2.49 Ma [1.50–3.60 Ma], restricted maternal gene flow could have allowed plastome haplotype frequencies to diverge substantially. The upper Yarlung Tsangpo valley and associated mountain systems may have further reinforced this separation by limiting seed-mediated connectivity between these two lineages. This contrasts with many wind-pollinated trees, where extensive pollen-mediated connectivity can markedly weaken spatial structure, including QXP conifers

showing evidence of efficient pollen flow across fragmented forests (Meng et al., 2007; Kling and Ackerly, 2021; Li et al., 2023; Tang et al., 2025). Our findings underscore that in these complex montane systems, physical barriers can impose strong constraints on seed-mediated gene flow. Such pronounced spatial isolation is often more apparent in uniparentally inherited markers than in biparental nuclear loci, consistent with expectations under pollen–seed dispersal asymmetry (Ennos, 1994; Meng et al., 2007). Therefore, when interpreting alpine phylogeographic structure, it is essential to explicitly consider potential discordance between seed-mediated colonization and pollen-mediated connectivity, because constraints on seed dispersal can play a disproportionately strong role in shaping pronounced maternal lineage structure (Kling and Ackerly, 2021; Xu et al., 2021). These plastome-based inferences therefore define the maternal, seed-mediated component of lineage history and provide a focused framework for future tests using population-level nuclear genomic data.

4.2. Pliocene–Quaternary lineage divergence and demographic dynamics

The four genetic clusters of *Rhodiola bupleuroides* (A–D) form a discrete, stepped geographic replacement pattern, consistent with the broader view that long-term tectonic and topographic evolution has structured lineage assembly and diversification across the QXP and its margins (Xing and Ree, 2017; Spicer et al., 2021; Zhu et al., 2026). Our molecular dating indicates that the onset of lineage divergence in *R. bupleuroides* occurred during the Pliocene, with the major maternal lineage splits extending into the Early Quaternary. This chronology is broadly consistent with continuing topographic and environmental reorganization of the QXP and its margins during the late Cenozoic (An et al., 2001; Ao et al., 2016; Miao et al., 2022). This temporal framework is further supported by recent regional syntheses showing that the present-day biogeographic structure of the QXP and the Himalaya–Hengduan region reflects the combined effects of Cenozoic topographic reorganization, climatic differentiation, and historical assembly of alpine floras (Spicer et al., 2020; Yu et al., 2024; Ahmad et al., 2025; Cao et al., 2025a). In particular, phylogenetic and floristic analyses have shown that the QXP contains distinct floristic subregions shaped by geographical distance, climatic gradients, topographic heterogeneity, and historical processes (Yu et al., 2024), while recent macroevolutionary evidence indicates that Cenozoic geoclimatic changes contributed to the asynchronous development of floristic endemism across different QXP subregions (Cao et al., 2025a). Geological syntheses further indicate that QXP uplift and plateau-margin relief development were spatially heterogeneous and multi-stage processes, accompanied by progressive river incision and drainage reorganization along the plateau margins (Wang et al., 2012; Ding et al., 2022; Cai et al., 2023). The estimated divergence of the major maternal lineages in *R. bupleuroides* from 3.88 Ma [2.83–5.31 Ma] to 2.49 Ma [1.50–3.60 Ma] therefore falls within a late Cenozoic interval during which canyon–ridge relief, major river systems, and plateau-margin topographic transitions were progressively established or intensified. This broad temporal concordance strengthens the interpretation that the mountain–canyon–river configuration provided a long-term spatial template for maternal lineage divergence in this seed-limited alpine herb. In this context, the maternal lineage history of *R. bupleuroides* fits a broader QXP biogeographic framework in which mountain building, river incision, habitat fragmentation, and climatic reorganization jointly shaped alpine lineage diversification (Ding et al., 2020; Cheng et al., 2024; Cao et al., 2025a). Similar patterns have also been reported from other mountain systems, where orogenesis and landscape fragmentation established spatial templates for

lineage divergence before later climatic oscillations further modified demographic dynamics (Antonelli et al., 2018; Smyčka et al., 2022).

We propose that late Cenozoic topographic evolution may have facilitated allopatric divergence via coupled processes of ecological opportunity (expansion into newly available alpine habitats) and increasing physical isolation as relief intensified and major drainage systems became more deeply incised (Mao et al., 2021; Ding et al., 2022; Rana et al., 2023). As the plateau landscape evolved, it may have increased both the availability and fragmentation of alpine habitats, creating opportunities for range shifts while also strengthening spatial isolation among populations (Spicer et al., 2020). In parallel, the growth of relief and associated fluvial incision along the eastern Tibetan margin likely promoted fragmentation of once more continuous ranges into geographically isolated units. These barriers would be expected to strongly reduce effective seed-mediated dispersal, thereby reinforcing lineage divergence once established (Cruzan and Hendrickson, 2020). While Cluster A represents the earliest-diverging maternal lineage along the eastern margin, the subsequent divergence of Clusters B (Hengduan Mountains), C (plateau interior), and D (southwestern margin) traces a stepped spatial lineage pattern broadly associated with the geomorphological evolution of the plateau.

Although Quaternary climate oscillations are frequently emphasized in temperate phylogeography, our molecular dating places the major maternal lineage divergences of *Rhodiola bupleuroides* within a Pliocene to Early Quaternary framework (Xing and Ree, 2017; Mao et al., 2021). This temporal pattern suggests that lineage formation began before the most recent demographic fluctuations inferred from the BSP analyses, whereas later climatic oscillations likely modulated population size changes within already differentiated maternal lineages. The demographic increase inferred for Cluster D around 1 Ma, together with the slight recent increase inferred for Cluster C, is consistent with lineage-specific demographic adjustment during Quaternary climatic fluctuations (Wang et al., 2009; Li et al., 2020b). These changes may reflect shifts in suitable habitat availability, local persistence, or range redistribution under changing climatic conditions (Hewitt, 2000). Taken together, our results support a two-stage scenario in which Pliocene to Early Quaternary topographic and environmental changes provided a spatial and ecological framework for maternal lineage structure, while later Quaternary climatic oscillations modulated lineage-specific demographic dynamics (Mao et al., 2021; Cao et al., 2025a).

4.3. Stronger IBD than IBE signals in plastome genetic structure

Our multidimensional analyses (Barrier tests, Mantel/partial Mantel, and dbrDA) support a hierarchical but joint pattern in *Rhodiola bupleuroides*, in which geographic isolation shows a stronger broad-scale association with plastome genetic structure, while environmental heterogeneity retains a detectable but relatively limited independent signal (Legendre et al., 2015; Capblancq and Forester, 2021; Quilodrán et al., 2025). This pattern fits the isolation-by-resistance framework for complex mountain landscapes, where deep valleys, steep elevational transitions, and river systems can increase landscape resistance and disproportionately restrict seed-mediated dispersal (Guan et al., 2024).

The genetic discontinuities identified here are geographically concordant with the plateau's "mountain-canyon-river" configuration, consistent with the view that major gorge-ridge systems can act as effective barriers to plant gene flow in the Hengduan-LRGR region (Qiao et al., 2022; Wambulwa et al., 2022). Together with the divergence-time estimates discussed above, this

spatial concordance indicates that the major topographic structures of the QXP and adjacent mountains have acted as persistent landscape-resistance features during the diversification history of *Rhodiola bupleuroides*. Most notably, the deeply incised Three Parallel Rivers canyon system (Barriers b and c) likely constitutes an important resistance feature for effective dispersal, contributing to the separation of Cluster B from other clusters and to finer sub-structuring via within-region fragmentation (Spear et al., 2010; Wambulwa et al., 2022). These findings are consistent with the isolation-by-resistance (IBR) framework, in which steep elevational discontinuities and major river-gorge systems can impose high landscape resistance, thereby limiting effective migration—especially for taxa with limited seed dispersal capacity (McRae, 2006; Spear et al., 2010; Cruzan and Hendrickson, 2020; Wambulwa et al., 2022). For maternally inherited plastomes, such resistance is expected to be particularly evident because recurrent seed-mediated migration, rather than occasional long-distance establishment, determines the extent of maternal genetic connectivity among populations. The stronger correlation of genetic distance with geographic distance ($r = 0.578$, $P = 0.0001$; Fig. 3B–D) supports the stronger broad-scale association of geographic isolation with plastome variation.

After controlling for geographic distance, environmental distance retained a significant but relatively limited independent association with plastome genetic differentiation ($r = 0.051$, $P = 0.0057$). This result indicates that environmental heterogeneity has a detectable independent contribution, although its effect size is much smaller than that of geographic distance, especially given ongoing debates about the performance of partial Mantel tests under complex spatial autocorrelation (Guillot and Rousset, 2013; Legendre et al., 2015; Quilodrán et al., 2025). The dbrDA results provided complementary evidence for this interpretation: after VIF filtering, MI, Bio8, Bio4, Bio2, and ELE were identified as the main environmental correlates of plastome genetic structure. These variables indicate that moisture availability, temperature-related gradients, and elevation-associated conditions are associated with local-scale differentiation, consistent with isolation by environment (IBE) processes acting alongside the stronger effect of geographic isolation (Wang and Bradburd, 2014; Capblancq and Forester, 2021). Thus, environmental heterogeneity is unlikely to be the primary driver of lineage structure, but it may have contributed to finer-scale differentiation among lineages occupying contrasting plateau and Hengduan habitats.

Ultimately, the spatial genetic patterns of *Rhodiola bupleuroides* are the product of a hierarchical interaction between topographic structures defining dispersal boundaries and environmental heterogeneity influencing colonization success (Manel and Holderegger, 2013; Cruzan and Hendrickson, 2020). This hierarchical mechanism is particularly evident in the Hengduan Mountains, where physical fragmentation and vertical habitat zonation together can maintain fine-scale maternal structure and provide the spatial background for lineage-specific plastome variation within a spatially compact but topographically complex region (Fu et al., 2022; Wambulwa et al., 2022).

4.4. Lineage-specific plastome variation in Hengduan and plateau habitats

The topographic barriers identified in Section 4.3 may have contributed to lineage-specific plastome variation by maintaining spatial separation among maternal lineages over time (McRae, 2006; Igea and Tanentzap, 2021). The contrast between the Hengduan Mountains (Cluster B) and the plateau interior (Cluster D) suggests that topographic isolation and habitat heterogeneity may have shaped different combinations of plastome diversity,

codon usage patterns, and demographic histories. Importantly, these interpretations are based on the combined evidence from haplotype diversity, nucleotide diversity, neutrality tests, demographic reconstruction, and plastome-level genomic features, rather than on haplotype network topology alone.

Cluster B, distributed in the core of the Hengduan Mountains, showed high haplotype richness, high nucleotide diversity, weak codon usage bias, and enrichment of complex SSR motifs. In the Hengduan Mountains, complex topography and microclimates can buffer macroclimatic extremes and promote long-term lineage persistence over long timescales (Finocchiaro et al., 2023; Keppel et al., 2024). This is reflected at the microstructural level: Cluster B showed a greater representation of complex SSR motifs—tri- and penta-nucleotide repeats—predominantly located within introns. Coupled with high ENC values (indicating weak codon bias), these patterns suggest that long-term lineage persistence and internal topographic dissection, such as the Three Parallel Rivers canyons, may have promoted fine-scale population subdivision and the retention of plastome sequence and microstructural variation within this lineage (Wambulwa et al., 2022; Chang et al., 2023). In this sense, the complex SSR profile of Cluster B is best understood as part of a broader high-diversity maternal lineage pattern in the Hengduan Mountains, where long-term persistence and internal landscape fragmentation may have jointly preserved plastome variation.

Cluster D, distributed mainly along the southwestern plateau margin and high-elevation interior, showed a different plastome pattern characterized by stronger codon usage bias, low nucleotide diversity, and reduced retention of complex SSR motifs. The negative association between ENC and elevation indicates that elevation-associated environmental gradients may contribute to codon usage variation within the sampled range. Under the translational selection framework, such a pattern is consistent with stronger functional constraint on plastome coding sequences, potentially maintained by purifying selection, particularly under energetic or physiological stress (Hershberg and Petrov, 2008; Plotkin and Kudla, 2011; Zhang et al., 2024a). However, stronger environmental pressure does not necessarily predict a larger effective population size (Leffler et al., 2012; Ellegren and Galtier, 2016). Environmental pressure reflects the strength of functional constraints or selective filtering, whereas N_e is shaped by demographic history, habitat fragmentation, lineage isolation, and stochastic processes (Corbett-Detig et al., 2015). Therefore, the low nucleotide diversity and demographic fluctuation observed in Cluster D are not inconsistent with stronger plastome sequence constraint. Codon usage bias can also be influenced by drift, linked selection, and lineage-specific demographic history. Thus, the Cluster D pattern is best interpreted as the combined outcome of local habitat conditions, restricted maternal gene flow, and lineage-specific demographic history, with adaptive selection remaining a hypothesis that requires further testing.

Together, these lineage-level contrasts suggest that topographic isolation and habitat heterogeneity have shaped different combinations of plastome diversity, microstructural variation, codon usage bias, and demographic history across the QXP. Rather than indicating sharply distinct adaptive regimes, these patterns point to lineage-specific plastome trajectories shaped by restricted maternal gene flow, local habitat conditions, and historical demography. This interpretation is consistent with the plastome-centered nature of the intraspecific dataset. The genus-level nuclear framework provides broader phylogenetic context for these environmental patterns, while population-level nuclear genomic data will be needed to test whether similar lineage-specific signals occur in biparentally inherited markers within *R. bupleuroides*.

4.5. Genus-level nuclear context for environmental structuring in *Rhodiola*

The genus-level nuclear analyses place the plastome-based environmental patterns of *Rhodiola bupleuroides* within a broader phylogenetic framework. In the 353-gene nuclear species tree of *Rhodiola*, the clearest phylogenetic signals were detected for Bio2, Bio3, and PET, whereas Bio4, BSAT, ALUM_SAT, and other variables showed marginal signals. This pattern indicates that temperature variability, evapotranspiration-related water demand, and some edaphic properties show phylogenetic structure across the sampled *Rhodiola* phylogeny. This broader pattern is consistent with the ecological setting of alpine *Rhodiola*, in which thermal heterogeneity, water availability, and soil conditions may jointly influence species distributions and lineage differentiation across the QXP and adjacent mountains (Wen et al., 2014; Zhang et al., 2014, 2025a).

The RF-distance analysis further highlighted a subset of candidate environment-related genes whose single-gene trees were relatively congruent with the 353-gene nuclear species tree (Robinson and Foulds, 1981). Among the 53 candidate genes recovered with GeneMiner2, nine genes showed normalized RF distance ≤ 0.6 , including *DNAJ1*, *DREB3*, *ICE1*, *LEA3*, *PAL1*, *PAL2*, *PAL3*, *PAL5*, and *ZAT10*. These genes are mainly associated with stress-response processes, including heat or protein-folding stress, cold response, dehydration tolerance, oxidative stress regulation, and phenylpropanoid-related pathways (Chinnusamy et al., 2003; Mittler et al., 2006; Sharma et al., 2019; Shiraku et al., 2022). These functional categories are ecologically relevant to the environmental dimensions highlighted by the phylogenetic signal analysis, particularly temperature variability, evapotranspiration-related water demand, and habitat stress (Li et al., 2020a; Zhang et al., 2024b). They also broadly echo the population-level plastome analyses of *Rhodiola bupleuroides*, in which moisture-, temperature-, and elevation-related gradients were identified as detectable environmental correlates of genetic differentiation. Notably, *PAL5* showed the lowest RF distance, followed by *DNAJ1*, suggesting relatively high topological congruence between these gene trees and the nuclear species tree. Together, these genus-level patterns provide a broader nuclear phylogenetic context for interpreting the plastome results and highlight environmental variables and candidate genes that warrant further examination in future population-level nuclear genomic studies of *R. bupleuroides*.

4.6. Methodological limitations and future directions

The scope of inference in this study is shaped by the marker system and analytical framework used here. The intraspecific analyses were based primarily on complete plastomes, which provide high-resolution information on maternal lineage history and seed-mediated genetic structure in angiosperms. This marker system is therefore particularly informative for detecting geography-associated maternal discontinuities in a seed-limited alpine herb. At the same time, plastome patterns represent the maternal component of lineage history and may differ from patterns recovered from biparentally inherited nuclear markers, especially under cytonuclear discordance caused by pollen-mediated gene flow or organellar history (Xu et al., 2021; Doyle, 2022; Fu et al., 2024). Population-level nuclear genomic data will therefore be important for testing whether the maternal structure recovered here is mirrored, weakened, or reorganized in the nuclear genome.

The landscape genetic analyses provide a statistical framework for evaluating the relative associations of geographic isolation and environmental heterogeneity with plastome genetic

differentiation (Quilodrán et al., 2025). Mantel tests, partial Mantel tests, and dbRDA consistently showed that geographic distance had a stronger association with plastome genetic structure than environmental distance, while some environmental variables retained detectable signals after geographic distance was included as the control matrix. These results support a hierarchical interpretation in which topographic fragmentation provides the dominant spatial template for maternal lineage structure, with environmental heterogeneity contributing additional but more limited associations. Further demographic modelling, resistance-surface analyses, and independent genomic datasets would help refine how historical dispersal limitation, habitat filtering, and lineage-specific demographic processes interacted through time (Spear et al., 2010; Epps and Keyghobadi, 2015).

The molecular dating results should also be interpreted as an approximate temporal framework for maternal lineage diversification. The divergence-time estimates were based on secondary calibration constraints and are influenced by taxon sampling, calibration uncertainty, clock-model assumptions, and HPD intervals (Warnock et al., 2015). For this reason, the inferred Pliocene to Early Quaternary timing is most useful for placing the major maternal lineage divergences within a broad late Cenozoic context, while precise links between individual splits and specific tectonic or climatic events remain hypotheses for future testing. The plastome variation analyses were designed to characterize lineage-associated genomic patterns and to generate testable hypotheses about their evolutionary significance. The concatenated-CDS Ka/Ks analysis provides an exploratory overview of broad coding-sequence divergence, rather than a formal gene-level test of positive selection (Hurst, 2002). Similarly, lineage differences in SSR abundance and motif composition reveal plastome micro-structural differentiation, but their functional significance remains to be evaluated. The genus-level RF analysis of candidate environment-related genes provides complementary nuclear phylogenetic context and highlights genes worthy of further investigation. Together, these considerations place the plastome-based findings in a defined interpretive framework and identify the nuclear genomic, demographic, and functional evidence needed to further evaluate the mechanisms proposed here. Given its occurrence in fragmented high-elevation habitats, the strong maternal lineage structure recovered here provides a genetic basis for recognizing lineage-level evolutionary units when considering the conservation value of this species (Tao et al., 2024).

5. Conclusions

By leveraging population-scale plastome sequencing, this study identifies *Rhodiola bupleuroides* as an informative case study for understanding alpine plant evolution. We demonstrate that while the plastome maintains structural stasis, its micro-scale sequence dynamism provides the resolution necessary to resolve seed-mediated maternal lineage structure in a highly dissected alpine landscape. Molecular dating placed the onset of intraspecific maternal lineage divergence at 3.88 Ma [2.83–5.31 Ma], with subsequent splits extending from the Pliocene into the Early Quaternary. This temporal framework suggests that late Cenozoic landscape reorganization provided an important spatial and ecological background for maternal lineage divergence, whereas later Quaternary climatic fluctuations further modulated lineage-specific demographic dynamics.

Our results establish a hierarchical pattern in which geographic isolation shows a stronger broad-scale association with plastome genetic structure than environmental heterogeneity, while moisture-, temperature-, and elevation-related environmental gradients retain detectable but relatively limited signals. Cluster B in the

Hengduan Mountains represents a high-diversity maternal lineage characterized by high chloroplast DNA diversity and enriched complex SSR motifs, whereas Cluster D shows stronger codon usage bias, lower nucleotide diversity, and reduced retention of complex SSR motifs. These contrasts are best interpreted as lineage-specific plastome variation shaped by restricted seed-mediated gene flow, local habitat conditions, and demographic history, rather than as direct evidence for sharply distinct selective regimes.

Consequently, our findings highlight the importance of topographic fragmentation and restricted seed-mediated dispersal in shaping maternal lineage differentiation in seed-limited alpine herbs, with the mountain–canyon–river configuration acting as a long-term spatial template for restricted maternal connectivity. The plastome patterns observed here are best understood as the combined outcome of landscape history, spatial isolation, environmental heterogeneity, and lineage-specific demographic processes. From a conservation perspective, although *Rhodiola bupleuroides* appears less directly affected by intensive harvesting than some medicinally exploited congeners, its occurrence in fragmented high-elevation habitats and its clearly differentiated maternal lineages indicate important evolutionary value for conservation. The four maternal lineages identified here therefore represent distinct evolutionary units that preserve different components of plastome diversity across the QXP and adjacent mountains, suggesting that future conservation planning should consider not only species-level occurrence but also lineage-level genetic distinctiveness and representative populations from different topographic regions. Future population-level nuclear genomic studies will be valuable for testing whether similar lineage-specific patterns are also present in biparentally inherited markers. Together, these results position *R. bupleuroides* as a plastome-based example of how topographic fragmentation can structure maternal lineage history in seed-limited alpine herbs on the QXP.

CRedit authorship contribution statement

Jia-Qing Hou: Writing – original draft, Visualization, Investigation, Formal analysis, Methodology. Yi Shi: Software, Visualization. Yong-Xiu Song: Visualization, Methodology. Xin-Yi Yu: Investigation. Xiao-Yu Jiang: Visualization. Fu-Sheng Yang: Writing – review & editing, Resources, Methodology, Supervision, Conceptualization. Yan Yu: Writing – review & editing, Writing – original draft, Resources, Methodology, Supervision, Conceptualization.

Data availability statement

The assembled chloroplast genome sequences have been deposited in the China National GeneBank Sequence Archive (CNSA) (<https://db.cngb.org/cnsa/>) under Project accession CNP0009022. Supplementary data to this article can be found online at Supplementary Tables.xlsx and Supplementary figures.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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